

ReLoop AI – Agentic E-Waste Recovery & Circularity Orchestrator

SOA Project Review-1



Under the Guidance of

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Problem Statement

Traditional e-waste disposal systems largely depend on users or collection centres manually deciding whether electronic devices should be repaired, reused, refurbished, or recycled. As the volume and variety of discarded electronics increases, this leads to:

- Premature disposal of repairable or reusable electronic devices.
- Lack of intelligent assessment of a device's condition and remaining useful life.
- Difficulty in identifying the most sustainable recovery option for each device.
- Limited coordination between users, repair/refurbishment centres, reuse partners, and recyclers.
- Loss of valuable components and materials that could be recovered and reused.
- Lack of proactive tracking and orchestration throughout the device's recovery journey.
- Limited intelligence in existing systems to make context-aware decisions between repair, reuse, refurbishment, parts recovery, and recycling.

Objectives

- Develop an Agentic AI-powered e-waste recovery system for intelligent and sustainable electronic-device management.
- Automatically assess the condition, repairability, reusability, and recovery potential of discarded electronic devices.
- Intelligently determine the most suitable pathway such as repair, reuse, refurbishment, parts recovery, or recycling.
- Automatically match devices with appropriate repair, refurbishment, reuse, and authorized recycling partners and coordinate the recovery process.
- Implement a scalable SOA and microservices-based architecture to improve e-waste recovery efficiency, reduce unnecessary disposal, and maximize material and device reuse.

Literature Survey

Authors	Title of the Paper	Methodology	Inferences	Limitation/Future work
Jano et al. (2026)	Closing the Loop: A Systematic Review of Artificial Intelligence in Circular E-Waste Management	- Reviewed 147 studies on AI in circular e-waste management. - Analysed AI applications across collection, classification, sorting, recovery and second-life use.	- AI can support multiple stages of the e-waste lifecycle. - Deep learning and computer vision are widely used, particularly for detection and classification. - AI has potential to support circular and closed-loop e-waste systems.	- Existing AI applications are largely fragmented across individual lifecycle stages. - Data availability, interoperability and standardization remain challenges.
Rajeev et al. (2025)	Advancing E-Waste Classification with Customizable YOLO Based Deep Learning Models	- Applied YOLO-based deep learning models for e-waste classification. - Used computer vision to identify different categories of electronic waste.	- Deep learning can improve automated identification and classification of e-waste. - Computer vision can reduce dependence on manual sorting.	- Primarily focuses on classification and identification. - Does not determine whether an individual device should be repaired, reused, refurbished or recycled. - Does not provide end-to-end recovery orchestration.
Oise & Konyeha (2025)	Environmental Impacts in E-Waste Management Using Deep Learning	- Used deep learning models including EfficientNet and MobileNet. - Classified different e-waste categories to support sorting and management.	- Deep learning can assist efficient e-waste classification and sorting. - Automated classification can contribute to improved environmental management.	- Mainly concentrates on classification and sorting. - Does not provide intelligent recovery-path selection.

Problem Gap Identification

- Existing e-waste systems mainly focus on individual tasks such as device classification, sorting, or recycling.
- Limited intelligent automation across the complete e-waste recovery lifecycle.
- High dependency on manual decisions for selecting repair, reuse, refurbishment, or recycling pathways.
- Lack of intelligent coordination between users, repair centres, refurbishers, reuse partners, and recyclers.
- Need for an integrated Agentic AI and SOA-based system that can assess devices, make recovery decisions, and orchestrate the complete circularity process.

Innovation, Creativity, and Novelty

- **Autonomous AI Recovery Workflow:** Multiple AI agents collaboratively assess electronic devices and determine the most suitable recovery pathway.
- **AI-Based Circularity Decision:** The system evaluates device condition and identifies whether it should be repaired, reused, refurbished, used for parts, or recycled.
- **Intelligent Partner Matching:** AI automatically matches devices with suitable repair centres, refurbishers, reuse organizations, or authorized recyclers based on device condition and requirements.
- **Proactive Recovery Orchestration:** The system automatically coordinates pickup, processing, status tracking, and notifications instead of relying entirely on manual intervention.
- **Integrated End-to-End Circularity:** Combines device assessment, recovery decision-making, partner matching, logistics, monitoring, and impact tracking in a single SOA-based microservices system.

Feasibility and Project Plan

Feasibility

- Uses existing AI tools, APIs, and open-source technologies for device assessment and intelligent decision-making.
- SOA and microservices enable independent, modular, and scalable development of the recovery workflow.
- Can be developed incrementally in modules/phases, starting with basic device registration and progressing to AI-driven recovery decisions.
- Suitable for real-world use by individuals, organizations, e-waste collection centres, refurbishers, and recyclers.

Project Plan

Phase 1

Requirements, e-waste workflow research & SOA architecture design

Phase 2

Device registration, user management & authentication services

Phase 3

AI agents for device assessment, recovery decision & partner matching

Phase 4

Logistics, recovery tracking, monitoring & notification services

Phase 5

SOA integration, Kafka event workflow, testing & deployment

Dataset Details

- **Device Information:** Dataset includes electronic-device details such as device type, brand/model, age, usage, and device category.
- **Condition & Fault Data:** Includes physical condition, functional status, reported faults, battery/screen condition, and available diagnostic information.
- **Recovery Attributes:** Contains recovery outcomes such as repair, reuse, refurbishment, parts recovery, and recycling.
- **Partner Information:** Includes partner type, specialization, location, service capabilities, availability, and recovery requirements.
- **AI Decision Support:** Dataset is used by the AI workflow for device assessment, recovery-path selection, and intelligent partner matching.

Front-End Implementation & Execution

- **User Interface:** Developed a web-based interface for users to register and submit electronic devices for recovery.
- **Device Registration:** Implemented forms to capture device type, brand/model, age, condition, faults, location, and other device information.
- **AI Assessment & Decision:** Front-end displays the device assessment, recovery scores, recommended recovery pathway, and decision information.
- **Recovery Tracking:** Provides users with visibility into recovery stages such as assessment, partner assignment, pickup, processing, and completion.
- **Backend Integration:** Front-end communicates with the project's backend services through REST APIs, enabling execution of the device recovery workflow.

Project Progress / Implementation Status

- **Architecture Implemented:** Established the SOA and microservices-based architecture with separate services for device management, AI assessment, decisions, partners, recovery, and notifications.
- **Core Workflow Implemented:** Implemented the initial workflow for device registration, device information management, and recovery processing.
- **AI Workflow Integrated:** Integrated the AI-based workflow for device assessment and circularity decision-making.
- **Front-End Implemented:** Developed and executed the user-facing screens for device submission, assessment, recovery decision, and workflow tracking.
- **Integration & Enhancement:** Progressing toward complete partner matching, recovery orchestration, Kafka event communication, monitoring, notifications, testing, and deployment.